AARHUS UNIVERSITY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Rev. 21/09/2025

Specifications of Bachelor of Science project for students at Aarhus University, Department of Electrical and Computer Engineering

Date	2025-09-21		
Project title	Satellite Image Super-Resolution Using Sentinel-1 and Sentinel-2 Data		
The project applies to students with a specialisation in (mark with X)			
Electrical Engineering		Computer Engineering	X
ECE supervisor (co-supervisor) info including name, email, mobile phone. *Note - main supervisor must be ECE VIP, Assistant Prof or more senior. PhD students, Postdocs and other experts (e.g. from industry, or from a different AU department) can co-supervise.*	Rune Hylsberg Jacobsen (email: rhj@ece.au.dk, mobile 4189 3252) Industry collaboration with Atla.ai – a spinout of Aarhus University – is an option for this project.		
Special demands to: - equipment - place - confidentiality	GPU-servers in NANLab.		

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Project description

Background/Motivations:

The ESA Copernicus program provides unprecedented access to satellite data through the Sentinel constellation, driving downstream innovation across numerous applications including environmental monitoring, agriculture, urban planning, and disaster response. However, practical applications are often limited by the spatial resolution constraints of satellite imagery.

Super-resolution technology aims at increasing image resolution through algorithmic means and has progressed significantly in recent years due to advances in computer vision and deep learning. This project proposes to leverage the complementary strengths of Sentinel-1 (Synthetic Aperture Radar, SAR) and Sentinel-2 (hyperspectral optical) data to achieve enhanced spatial resolution beyond the native capabilities of individual sensors.

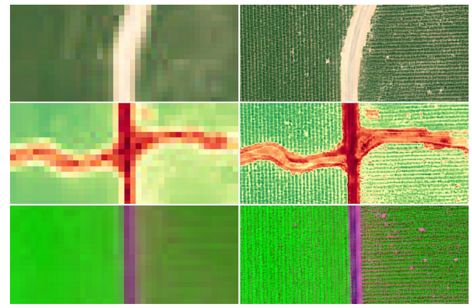

Figure 1: Example of 12-band 10x single image super-resolution for Sentinel-2 [1]

Objectives and methodology

The aim is to develop and evaluate a multi-modal deep learning approach for satellite image super-resolution by fusing Sentinel-1 SAR data with Sentinel-2 optical imagery to generate high-resolution multispectral images. This involves implementing convolutional neural network architectures for multi-sensor fusion, assessing the contribution of SAR textural information to optical image enhancement, and comparing the performance of single-modal versus multi-modal super-resolution methods.

The methodology combines traditional remote sensing data processing with modern deep learning architectures to leverage the complementary strengths of SAR and optical sensors. Data sources include Sentinel-2: Multispectral optical imagery (10 m resolution for visible/NIR bands, 20 m for SWIR bands), Sentinel-1: C-band SAR imagery in Interferometric Wide (IW) mode (~10 m resolution).

Outline of the methodology
Phase 1 - Data preprocessing: 1) Co-registration of Sentinel-1 and Sentinel-2 imagery.2) Radiometric calibration and atmospheric correction. 3) Data augmentation techniques for training dataset expansion.
Phase 2 - Model development: 1) Implementation of baseline single-modal CNN super-resolution models. 2) Development of multi-modal fusion architectures: i) Dual-branch CNN with early/late fusion strategies, ii) Attention-based feature fusion mechanisms, iii) GAN approaches for texture enhancement.
Phase 3 - Training and validation: 1) Loss function design combining pixel-wise and perceptual losses, 2) Training on degraded image pairs with known ground truth, 3) Cross-validation using geographically separated test areas.
Phase 4 - Performance evaluation: 1) Quantitative metrics: PSNR, SSIM, spectral fidelity measures. 2) Visual quality assessment and edge preservation analysis. 3) Comparison with state-of-the-art single-modal methods.

Learning outcomes:

- Technical Skills: Multi-modal remote sensing data processing and analysis, deep learning frameworks, CNN architecture and training methodologies, satellite image preprocessing, co-registration, and calibration techniques, performance evaluation metrics for super-resolution applications.
- Domain Knowledge: Understanding of SAR and optical satellite sensor characteristics, image fusion and multi-sensor remote sensing, applications of super-resolution in Earth observation, Copernicus program data.

References:

[1] Yosef Akhtman, Sentinel-2 Deep Resolution 3.0, https://medium.com/@ya_71389/sentinel-2-deep-resolution-3-0-c71a601a2253